

BUGEMA UNIVERSITY

Department of Computing and Informatics

Research Methods in Computing | Assignment 11

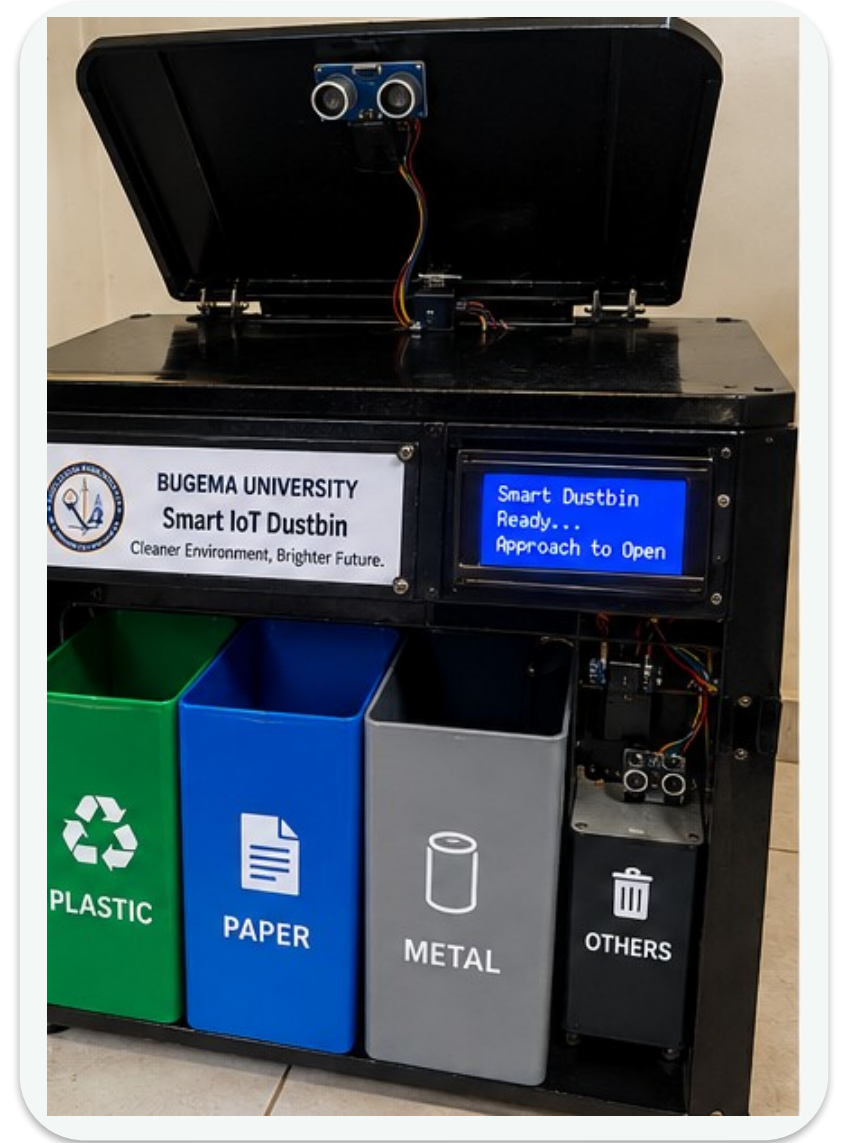
Design and Implementation of a **Smart IoT Dustbin** with Automatic Sorting

Touchless lid control, waste sorting
and IoT monitoring.

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24/BSE/BU/R/0001

Individual proposal | September 2026



AI-generated assembly concept.

Problem statement

A proposed campus recycling point; local needs will be investigated.

Mixed bins cannot separate waste or show when they are full.
Touch-operated lids require contact and may be unhygienic.
At a proposed Bugema recycling point, local needs must first be established.
This study will test a touchless bin that sorts selected items, rejects unsupported items, and reports fill levels safely.



AI-generated context image.
Not a photograph of Bugema University.

Research question: Can it open, sort and report bin status reliably?

Four specific SMART objectives

Ten-week plan; starts after approval and resource confirmation.

- | | | |
|----|---------------|---|
| 01 | BY WEEK
2 | Interview 8 students + 2 cleaning staff; observe disposal; define at least 10 prioritised requirements. |
| 02 | BY WEEK
5 | Build 1 guarded, four-compartment prototype: touchless lid, sensing and mechanical sorting. |
| 03 | BY WEEK
8 | Train and validate a classifier on $\geq 1,200$ images; connect fill-level alerts to an IoT dashboard. |
| 04 | BY WEEK
10 | Complete 120 sorting trials, 40 approach tests, 20 fault tests and a 10-person supervised pilot. |

Components and their roles

Close-up views from the proposed assembled dustbin concept.



Arduino UNO

Sensor and actuator control



Load cell / HX711

Item-weight measurement



HC-SR04 sensor

Approach detection



ESP32

Wi-Fi / IoT updates



Servo motor

Lid opening and waste routing



LCD display

Local system-status messages



IR sensor

Object-presence detection



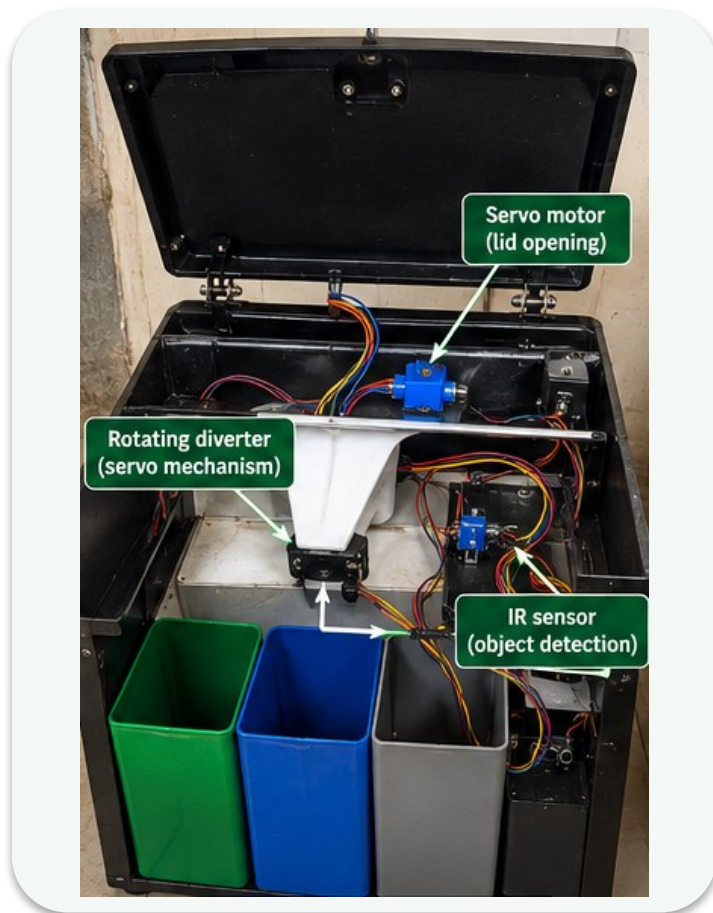
Power + four bins

Power supply and waste storage

Supplied AI-generated component illustrations; the written component list is retained.

Intended technological solution

Presence detection, inspection, sorting and connected monitoring.



AI-generated assembly concept.

- 1 Sense presence and open**
Ultrasonic detection requests lid opening.
- 2 Inspect the item**
IR, load-cell and optical inspection.
- 3 Direct it to a compartment**
Plastic, paper, metal or others.
- 4 Display and report status**
LCD feedback; ESP32 sends IoT updates.

Local control stays active during network loss.

References

Source entries retained from the supplied presentation.

[1] World Bank (2026)

What a Waste 3.0

[2] Bin-e (n.d.)

Smart waste bin

[3] Espressif Systems

ESP32 product information

[4] Arduino

UNO Rev3

[5] TensorFlow

Transfer learning and fine-tuning

[6] OASIS

MQTT Version 5.0

[7] HC-SR04

Ultrasonic sensor module documentation

[8] Micro servo motor

Product documentation

[9] IR obstacle sensor

Product documentation

[10] HX711

Load cell amplifier documentation

Entries [7–10] had no exact publisher or URL in the source deck.